SOLUTIONS: STABILITY AND CONTROL OF NONLINEAR SYSTEMS
MSC IN CONTROL & OPTIMISATION

1. Exercise

a) The vector field $f_2(x_1,x_2)$ is polynomial with respect to both variables. The function $f_1(x_1,x_2)$ is linear in x_2 and piecewise linear in x_1. Hence $f(x)$ is a locally Lipschitz function and by Caratheodory's theorem solutions exist and are unique (over their maximal time-interval of definition). [2]

b) The nullclines of the vector-field can be computed as shown below:

$$\mathscr{N}_1 = \{x : x_2 = |x_1+1| - |x_1-1| - x_1\}.$$

Notice that the nullcline corresponds to the image of a piecewise linear function of x_1, with slope $-1, 1$ and -1 for x_1 in the intervals $(-\infty,-1)$, $(-1,1)$ and $(1,+\infty)$ respectively. For the second nullcline we have:

$$\mathscr{N}_2 = \{x : x_1 = x_2^3 - 2x_2\}.$$

This is a cubic with negative slope (-2) passing through the origin.

c) The nullclines are sketched in FIg. 1.1. There are 4 regions, with orientations as indicated in the figure. [3]

d) None of the regions is forward invariant since each region has a portion of the boundary with outgoing arrows, showing that the vector-field pushes solutions to cross those boundaries. [2]

e) Nullclines can be seen to have a single intersection at the origin. This is the unique equilibrium of the system. [2]

f) Computing the Jacobian of $f(x)$ yields (where defined):

$$\frac{\partial f}{\partial x} = \begin{bmatrix} \text{sign}(x_1+1) - \text{sign}(x_1-1) - 1 & -1 \\ 1 & 2 - 3x_2^2 \end{bmatrix}.$$

Notice that the Jacobian is well defined for $x = 0$ and that the system is $\mathscr{C}^1$ in a neighborhood of 0. In particular:

$$A = \left.\frac{\partial f}{\partial x}\right|_{x=0} = \begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix}.$$

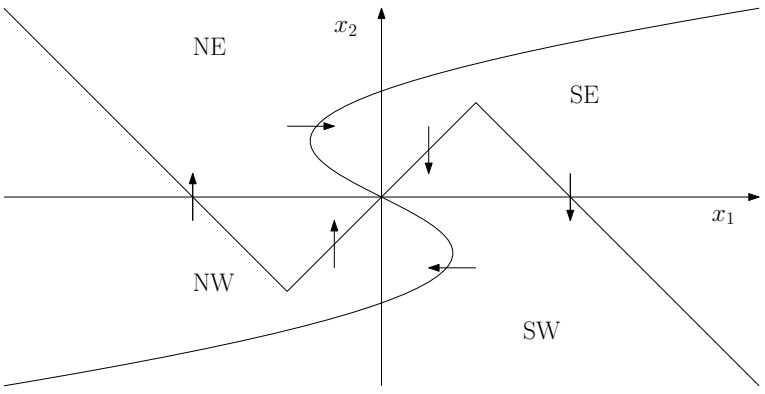


Figure 1.1 Regions and nullclines

The characteristic polynomial of A is $\chi(s) = s^2 - 3s + 3$ with complex conjugate roots in $3/2 \pm \sqrt{3}/2$. These eigenvalues correspond to an unstable focus. [3]

g) For $x_1 = 4$ we see $\dot{x}_1 = -x_2 + 2 - 4 = -x_2 - 2 \leq 0$ for all $|x_2| \leq 2$. Hence soludions move towards the interior of the box. A similar argument applies to $x_1 = -4$.
For $x_2 = 2$ we see $\dot{x}_2 = -8 + 4 + x_1 = -4 + x_1 \leq 0$ for all $|x_1| \leq 4$, hence also along this portion of the boundary solutions are pushed towards the interior of the box. A symmetric argument applies for $x_2 = -2$. Since along the boundary of the box solutions are pushed towards its interior we see that the box is forward invariant. [3]

h) Given that we have an unstable focus at the origin it is possible to consider a sufficiently small open disc D centered around the origin so that the set $B \backslash D$ is forward invariant. Indeed, for small x_1 we have: $\dot{x}_1 = -x_2 + x_1$. Taking derivatives of $x_1^2 + x_2^2$ along solutions yields:

$$2x_1\dot{x}_1 + 2x_2\dot{x}_2 = 2x_1^2 - 2x_1x_2 + 2x_2(-x_2^3 + 2x_2 + x_1) = 2x_1^2 + 4x_2^2 - 2x_2^4$$

which is positive definite in a neighborhood of 0. This shows that $\{x : x_1^2 + x_2^2 \geq M\}$ is forward invariant for M sufficiently small. Hence, $B \cap \{x : x_1^2 + x_2^2 \geq M\}$ is also forward invariant (notice this is the original box B with a disc-shaped hole). This set is compact and does not contain equilibria, hence by Poincaré-Bendixson criterion $\omega(x_0)$ is a closed curve for all x_0 in the set. In particular the initial condition $[1,1]'$ belongs to the set (for $M < 2$ which can be assumed w.l.o.g.) and therefore the ω-limit set is a closed curve. [3]

2. Exercise

a) We choose $x_1 = y_1$ for the scalar system Σ_1. Accordingly:

$$\dot{x}_1 = -2x_1 + u_1, \qquad y_1 = x_1.$$

[2]

We choose $x_2 = [z, \dot{z}]' = [(x_2)_1, (x_2)_2]'$ for the two-dimensional system Σ_2. Accordingly:

$$\dot{x}_2 = \begin{bmatrix} (x_2)_2 \\ -(x_2)_2^5 - \sin((x_2)_1) + u_2 \end{bmatrix}, \qquad y_2 = (x_2)_2.$$

[2]

b) We pick the storage function $S_1(x_1) = x_1^2/2$. Taking derivatives along solutions yields:

$$\dot{S}_1 = x_1\dot{x}_1 = -2x_1^2 + x_1u_1 \leq h_1(x_1)u_1.$$

This proves passivity (strict output passivity) of system Σ_1. [2]

c) Notice that:

$$y_2u_2 = \dot{z}[\ddot{z} + \dot{z}^5 + \sin(z)] = \frac{d}{dt}\left[\frac{\dot{z}^2}{2} - \cos(z)\right] + \dot{z}^6$$

$$\geq \frac{d}{dt}\left[\frac{\dot{z}^2}{2} - \cos(z)\right].$$

Hence, choosing the storage function $S_2(x_2) = \frac{(x_2)_2^2}{2} - \cos((x_2)_1)$ we have

$$\dot{S}_2 = h_2(x_2)u_2 - (x_2)_2^6 \leq h_2(x_2)u_2,$$

which shows passivity (strict output passivity) of subsystem Σ_2. [4]

d) It Is well know that parallel interconnections of passive systems are passive. In particular letting $x = [x_1, x_2']'$ there exists a (non-negative) storage function $S(x) = S_1(x_1) + S_2(x_2)$ such that the following inequality is fulfilled:

$$S(\varphi(t, x_0, u)) - S(x_0) \leq \int_0^t u(t)y(t)\,dt.$$

for all t. Taking $t = T$ yields:

$$0 = S(x(T)) - S(x_0) \leq \int_0^T u(t)y(t)\,dt,$$

where the first inequality follows since the solution is periodic and the second by passivity. [4]

e) We choose the Lyapunov function $V(x_2) = \frac{(x_2)_2^2}{2} + \frac{k}{2}(x_2)_1^2 - \cos((x_2)_1) + 1$. This is positive definite. In fact $1 - \cos((x_2)_1) \geq 0$. Moreover, $V(x_2)$ is lower-bounded by a positive definite quadratic form. Hence it is radially unbounded. Taking derivatives along solutions of the closed-loop system yields:

$$\dot{V} = -(x_2)_2^6 \leq 0.$$

This is negative semi-definite, and $\text{Ker}[\dot{V}] = \{x_2 : (x_2)_2 = 0\}$. Any invariant set contained in $\text{Ker}[\dot{V}]$ is also included in $\{x_2 : (x_2)_2 = 0$, and $-\sin((x_2)_1) - k(x_2)_1 - (x_2)_2^5 = 0\} = \{0\}$. Hence, all the assumptions of the global Lasalle's invariance principle are fulfilled and the origin is a GAS equilibrium. [6]

3. Exercise

a) Subsystem Σ_1 describes the evolution of the x_1 variable. Its input is u_1 and output y_1. The following is the associated state-space representation:

$$\dot{x}_1 = -g_1(2x_1) + g_1(u_1), \qquad y_1 = x_1.$$

Similarly, subsystem Σ_2 describes the evolution of the x_2 variable. Its input is u_2, output y_2 and the associated state-space representation:

$$\dot{x}_2 = -g_2(2x_2) + g_2(u_2), \qquad y_2 = x_2.$$

The original system can be seen as the feedback interconnection of Σ_1 and Σ_2 described by the following identifications: $u_1 = y_2$ and $u_2 = y_1$. [4]

b) For a generic scalar system of equations $\dot{x} = -g(2x) + g(u)$ with $g \in \mathscr{G}$ we may choose $V(x) = x^2/2$ as a candidate Lyapunov function. We see that:

$$\dot{V} = x(-g(2x) + g(u)) = -xg(2x) + xg(u) \leq -|x|g(2|x|) + |x|g(|u|).$$

In particular, for any $\varepsilon > 0$ and considering $x > 0$ yields

$$|x| \geq (1+\varepsilon)|u|/2 \Rightarrow \dot{V} \leq |x|[-g(2|x|) + g(2|x|/(1+\varepsilon))] < 0.$$

A similar inequality applies to the case $x < 0$. Hence $V(x)$ is an ISS Lyapunov function of type 2 and the system is Input-to-State Stable with gain $\gamma(r) = \frac{(1+\varepsilon)}{2} r$. Notice that $\gamma(r) \to r/2$ as $\varepsilon \to 0$. [6]

c) The loop gain of the interconnection is $\gamma_1(\gamma_2(r)) = \frac{(1+\varepsilon)^2}{4} r$. For ε sufficiently small this is less than the identity (in fact it approaches $\frac{1}{4} r$ as $\varepsilon \to 0$. By the small gain theorem the interconnection is globally asymptotically stable at the origin. [2]

d) The Input-Output Asymptotic Gain of the series interconnection can be obtained simply by composition of the I-S gains previously computed. In fact for individual subsystems we have:

$$\limsup_{t\to+\infty} |y_i(t)| = \limsup_{t\to+\infty} |x_i(t)| \leq \gamma_i(\limsup_{t\to+\infty} |u_i(t)|).$$

Considering their interconnection:

$$\limsup_{t\to+\infty} |y(t)| = \limsup_{t\to+\infty} |y_n(t)| \leq \gamma_n(\limsup_{t\to+\infty} |u_n(t)|) =$$

$$\gamma_n(\limsup_{t\to+\infty} |y_{n-1}(t)|) \leq \gamma_n(\gamma_{n-1}(\limsup_{t\to+\infty} |u_{n-1}(t)|)) =$$

$$\vdots$$

$$\gamma_n(\gamma_{n-1}(\ldots\gamma_2(\limsup_{t\to+\infty} |y_1(t)|)\ldots)) \leq \gamma_n(\gamma_{n-1}(\ldots\gamma_1(|u(t)|)\ldots)).$$

Therefore $\gamma(r) = \gamma_n(\gamma_{n-1}(\ldots\gamma_1(r)\ldots))$. In particular for this specific case $\gamma(r) = \frac{(1+\varepsilon)^n}{2^n} r$. [4]

e) For all feedback gains K with $|K| < 2^n$ the resulting loop gain is less than unity (choosing ε sufficiently small, in fact: $K\gamma(r) = K(1+\varepsilon)^n/(2^n)r \to K/(2^n)r$ as $\varepsilon \to 0$. As a consequence, there exists a sufficiently small ε so that $K\gamma(r) < r$ (for positive r). [4]

4. Exercise

a) To ascertain the relative degree we differentiate the output along solutions of the system:

$$\dot{y} = \dot{x}_1 = \arctan(x_1) + u(\alpha + \cos(x_2)).$$

Specifically the coefficient of u, equals $\alpha + \cos(x_2)$. Several cases arise as a function of α.

i) $|\alpha| > 1$, implies $\alpha + \cos(x_2) \neq 0$ for all $x \in \mathbb{R}^3$. Hence the relative degree is well defined globally and equal to 1;

ii) $\alpha \in (-1, 1]$ implies $\alpha + \cos(0) = \alpha + 1 \neq 0$ in a neighborhood of $x = 0$. However, the equation $\alpha + \cos(x_2) = 0$ admits solutions for $x_2 \neq 0$. Hence the relative degree is well defined locally and equal to 1;

iii) $\alpha = -1$ implies $\alpha + \cos(0) = 0$. Hence the relative degree is undefined.

[4]

b) The linearizing feedback is:

$$u = \frac{v - \arctan(x_1)}{\alpha + \cos(x_2)}.$$

Accordingly, for the closed loop system $\dot{x}_1 = v$. [3]

c) Any feedback of the form $v = kx_1$ with $k < 0$ would globally asymptotically stabilize the I-O dynamics around $y = 0$. [2]

d) We choose the following coordinates $\xi = x_1$, $\zeta_1 = x_2 - x_1$ and $\zeta_2 = x_3$. With this choice we get:

$$\begin{aligned} \dot{\xi} &= v \\ \dot{\zeta}_1 &= -\zeta_1 + \zeta_2 \\ \dot{\zeta}_2 &= -\zeta_1 - \zeta_2 + y^3 \end{aligned} \tag{4.1}$$

$$y = \xi$$

Therefore, the equations for ζ_1 and ζ_2, which are forced by y and independent of u, are the internal dynamics. [4]

e) The internal dynamics are of the form:

$$\dot{\zeta} = A\zeta + \begin{bmatrix} 0 \\ 1 \end{bmatrix} y^3$$

where y is the exogenous input signal, and A is a 2×2 Hurwitz matrix. Hence, the dynamics are ISS. This can be shown directly using the explicit solution formula for linear systems, or by means of ISS-Lyapunov functions, i.e. $V(\zeta) = \zeta^T \zeta$. [3]

f) When the relative degree is defined globally, we get a closed-loop system which is globally defined. Due to ISS of the internal dynamics and the cascaded nature induced by the normal form, a G.A.S. closed-loop system ensues. When the relative degree is defined locally, viz. for $\alpha \in (-1, 1]$, the closed-loop system is only defined locally and therefore we have a locally asymptotically stable closed-loop system. The proposed feedback is unsuitable for $\alpha = -1$, as it entails division by 0 at the equilibrium (infinite control actions near equilibrium).

[4]